Analysis of Algorithms
Lab 06

1. Solve the following recurrences using the Master Theorem:
   1.1. T(n) = 4 T(n/2) + n, T(1) = 1
   1.2. T(n) = 4 T(n/2) + n^2, T(1) = 1
   1.3. T(n) = 4 T(n/2) + n^3, T(1) = 1

2. Implement the Merge Sort algorithm.
3. Use your implementations and experiment to check if the merge sort is or is not a stable sorting algorithm. Provide and example.
4. Implement the Quick Sort algorithm and test using random generated lists.
5. Apply empirical analysis on the quick sort. Does it match with the mathematical analysis?

Submission

Upload your solutions as a PDF file. In case of coding assignments, upload to GitHub and include the link in the PDF file. Your submission should include screenshots of the execution of your code when applicable.